PN7150 NFC

U1
PN7150B0HN_C11002E-PN7150B0HN_C11002E

+3V3
R1 10K
R2 10K
SDA
SCL

RXP 16
RXN 15
ADR0 1
ADR1 3
SCL 7
SDA 5
27.12Mhz
Y1
C3 10pF
C4 10pF
VEN 10
GND

VDDD 30
VDD 29
VDDA 28
VDD(TX) 14
VBAT 26
VBAT1 13
VBAT2 12
VDD(TX_IN) 22
VDD(MID) 17
VDD(PAD) 6
TX1 21
TX2 18
CLK_REQ 40
IRQ 8
VSS(TX) 19
VSS(PAD) 4
VSS

C5 1uF
C6 1uF
C7 100nF
C8 1uF
C9 100nF
+3V3
GND

RXP
RXN
I2CADR0
I2CADR1
I2CSCL
I2CSDA
NFC_CLK_XTAL1
NFC_CLK_XTAL2
VEN

TX1
TX2
CLK
IRQ

LEDS & BUTTONS

SW1
RESET
RST
GND

GPIO
SW2
GPIO1
GND

R13 1K
D3
LED
LED1
GND

R12 1K
D2
LED
ESP_LED
GND

ANTENA MACHING CIRCUIT

RXP
C23 1nF/16V
L1 160nH
R15 2k2
TX1
100pF/50V
C27
C25 680pF
120pF
C30
R18 3.24R
L3
L
C26 680pF
GND
C31 120pF
R19 3.24R
TX2
L2 160nH
R16 2k2
100pF/50V
C28
RXN
C24 1nF/16V

FLASH

+3V3
10K R20
QSPI_CS
QSPI_SCK
C33 100nF
GND

U6
W25Q16JV
CS
CLK
DI(IO0)
DO(IO1)
IO2
IO3
VCC
GND
EP

QSPI_DATA0
QSPI_DATA1
QSPI_DATA2
QSPI_DATA3

MICROCONTROLLER

U2
+1V2
+3V3
DVDD DVDD VREG_VOUT VREG_IN IOVDD IOVDD IOVDD IOVDD IOVDD IOVDD USB_VDD ADC_AVDD

USB_DP
USB_DM
RD+
RD-
R9 27
R10 27
D+
D-

QSPI_CS
QSPI_DATA0
QSPI_DATA1
QSPI_DATA2
QSPI_DATA3
QSPI_SCK
QSPI_SS
QSPI_SD0
QSPI_SD1
QSPI_SD2
QSPI_SD3
QSPI_SCLK

GPIO0 GPIO1 GPIO2 GPIO3 GPIO4 GPIO5 GPIO6 GPIO7 GPIO8 GPIO9 GPIO10 GPIO11 GPIO12 GPIO13 GPIO14 GPIO15 GPIO16 GPIO17 GPIO18 GPIO19 GPIO20 GPIO21 GPIO22 GPIO23 GPIO24 GPIO25
ESP_RESET
VEN
IRQ
CIPO_TX
CS_RX
ACK_CTS
COPI_RTS
SDA
SCL
SCK
GPIO1
IN_B
IN_A
ESP_GPIO0

GPIO26_ADC0 GPIO27_ADC1 GPIO28_ADC2 GPIO29_ADC3
B_ADC
LED1

Raspberry Pi
RP2040

XIN
XOUT
Y2 12Mhz
1K R3
C1 27pF
C2 27pF
RUN
RST
SWCLK
SWDIO
SWCLK
SWD
TESTEN
GND

JP1
BOOT
QSPI_CS
GND

+3V3
C10 1uF
C12 100nF
C13 100nF
C15 100nF
C17 100nF
C19 100nF
C20 100nF
C21 100nF
GND

+1V2
C14 1uF
C16 100nF
C18 100nF
GND

WIFI & BLE NINA

+3V3
R17 10K
ESP_RESET
U5
ESP32-WROOM-32D
EN
VDD
SENSOR_VP
SENSOR_VN
SDO/SD0
SDI/SD1
SHD/SD2
SWP/SD3
SCK/CLK
SCS/CMD
GND

IO0 TXD0/IO1 IO2 RXD0/IO3 IO4 IO5 IO12 IO13 IO14 IO15 IO16 IO17 IO18 IO19 IO21 IO22 IO23 IO25 IO26 IO27 IO32 IO33 IO34 IO35
ESP_GPIO0
CS_RX
CIPO_TX
CS_RX
COPI_RTS
SCK
CIPO_TX
ESP_LED
ACK_CTS

+3V3
C32 100nF
GND

COIL

VBUS
+BATT
SW4A SOURCE MAG

U7
TC4424
NC NC
IN_A IN_A OUT_A OUTA
GND VDD
IN_B IN_B OUT_B OUTB
C34 10uf
GND

OUTA
OUTB
J3 MAGSPOOF

REGULATOR AND BATTERY

P1
USB_C_Plug_USB2.0
VBUS
CC
VCONN
D-
D+
SHIELD
GND
F1 0.5A
R4 5k1
R5 5k1
S1
A1

+BATT
B_ADC
R6 100K
R7 100K
GND

VBUS
U3
MCP73831-3-OT
VDD VSS PROG STAT VBAT
R8 5.1K
C11 10uF
R11 1K
D1
LED

+BATT
J1
Battery

VBUS
+BATT
Q1
DMG3415U-7
D4
MBR120
R14 100K
ON
SW3
C22 10uF
U4
AP2112K-3.3
VIN VOUT EN GND
C29 10uF
+3V3

ADDRESS SELECT

R21 100K
ADR0
R22 100K
ADR1
GND

H1 MountingHole
H2 MountingHole
H3 MountingHole
H4 MountingHole

Andres Sabas
Electronic Cats
Sheet: /
File: working.kicad_sch

Title: Bomber Cat

Size: A3	Date: 2023-01-06	**Rev: 1.2**
KiCad E.D.A. kicad-cli 7.0.6-7.0.6-ubuntu22.04.1		Id: 1/1